

Section: Particles and Waves

Topic: Refraction of Light

Light passes from air into a transparent material.

Angle of incidence in air: $i = 50^\circ$

Angle of refraction in the material: $r = 30^\circ$

(a)

Calculate the refractive index of the material.

Use Snell’s Law:

$$n = \frac{\sin i}{\sin r} = \frac{\sin 50^\circ}{\sin 30^\circ} = \frac{0.766}{0.500} = 1.53$$

 *Revision tip:*

Snell’s Law:

$$n = \frac{\sin i}{\sin r}$$

Use this when light **enters a denser medium** and slows down.

(b)

The speed of light in air is $3.0 \times 10^8 \text{ m/s}$.

Calculate the speed of light in the material.

Use:

$$n = \frac{c}{v} \Rightarrow v = \frac{c}{n} = \frac{3.0 \times 10^8}{1.53} = 1.96 \times 10^8 \text{ m/s}$$

 *Revision tip:*

Light slows down in denser materials.

$n > 1$ means $v < c$